R4 Newton's third law

Name:

Class:

Date:

Time: **397 minutes**

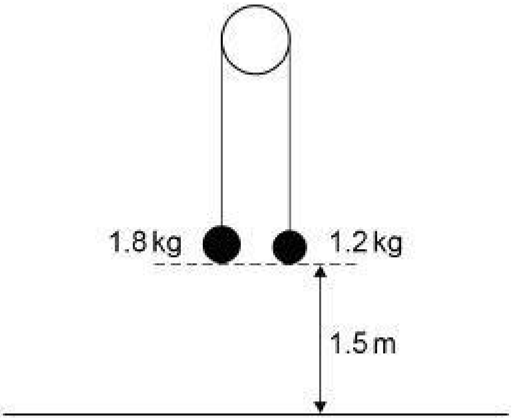
Marks: **331 marks**

Comments:

Q1.

In this question use g = 9.8 m s^{-2}

Two particles, of mass 1.8 kg and 1.2 kg, are connected by a light, inextensible string over a smooth peg.



(a) Initially the particles are held at rest 1.5 m above horizontal ground and the string between them is taut.

The particles are released from rest.

Find the time taken for the 1.8 kg particle to reach the ground.

(5)

(b) State one assumption you have made in answering part (a).

(1)

(Total 6 marks)

Q2.

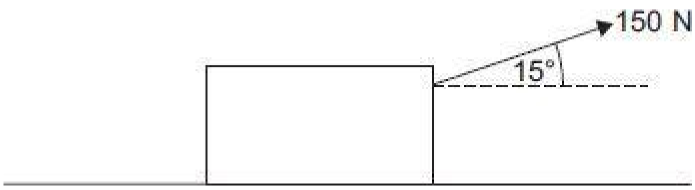
In this question use g = 9.8 m s^{-2}

A boy attempts to move a wooden crate of mass 20 kg along horizontal ground. The coefficient of friction between the crate and the ground is 0.85

(a) The boy applies a horizontal force of 150 N. Show that the crate remains stationary.

(3)

(b) Instead, the boy uses a handle to pull the crate forward. He exerts a force of 150 N, at an angle of 15° above the horizontal, as shown in the diagram.



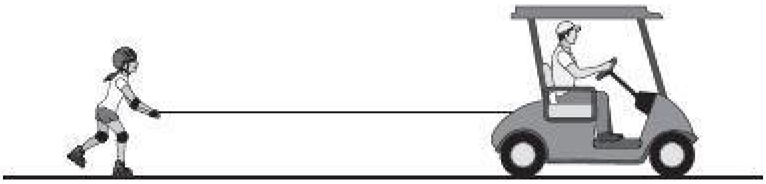
Determine whether the crate remains stationary.

Fully justify your answer.

(5)
(Total 8 marks)

Q3.

A buggy is pulling a roller-skater, in a straight line along a horizontal road, by means of a connecting rope as shown in the diagram.



The combined mass of the buggy and driver is 410 kg
A driving force of 300 N and a total resistance force of 140 N act on the buggy.

The mass of the roller-skater is 72 kg
A total resistance force of R newtons acts on the roller-skater.

The buggy and the roller-skater have an acceleration of 0.2 m s^{-2}

(a) (i) Find R.

(3)

(ii) Find the tension in the rope.

(3)

(b) State a necessary assumption that you have made.

(1)

(c) The roller-skater releases the rope at a point A, when she reaches a speed of 6 m s^{-1}

She continues to move forward, experiencing the same resistance force.

The driver notices a change in motion of the buggy, and brings it to rest at a distance of 20 m from A.

(i) Determine whether the roller-skater will stop before reaching the stationary buggy.

Fully justify your answer.

(5)

(ii) Explain the change in motion that the driver noticed.

(2)
(Total 14 marks)

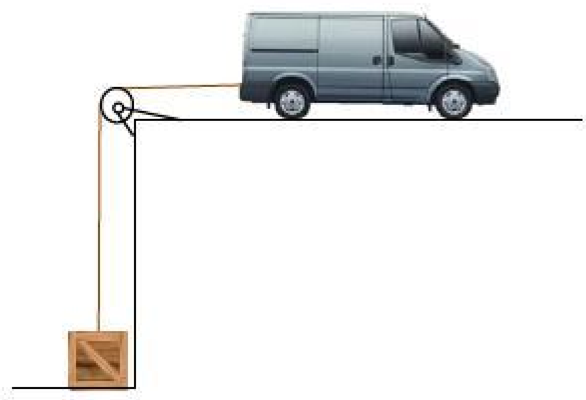
Q4.

In this question use g = 9.8 m s^{-2}, giving your final answers to an appropriate

degree of accuracy.

A van of mass 1300 kg and a crate of mass 300 kg are connected by a light inextensible rope.
The rope passes over a light smooth pulley, as shown in the diagram.

The rope between the pulley and the van is horizontal.



Initially, the van is at rest and the crate rests on the lower level. The rope is taut.
The van moves away from the pulley to lift the crate from the lower level.
The van's engine produces a constant driving force of 5000 N.
A constant resistance force of magnitude 780 N acts on the van.
Assume there is no resistance force acting on the crate.

(a) (i) Draw a diagram to show the forces acting on the crate while it is being lifted.

(1)

(ii) Draw a diagram to show the forces acting on the van while the crate is being lifted.

(1)

(b) Show that the acceleration of the van is 0.80 m s^{-2}

(4)

(c) Find the tension in the rope.

(2)

(d) Suggest how the assumption of a constant resistance force could be refined to produce a better model.

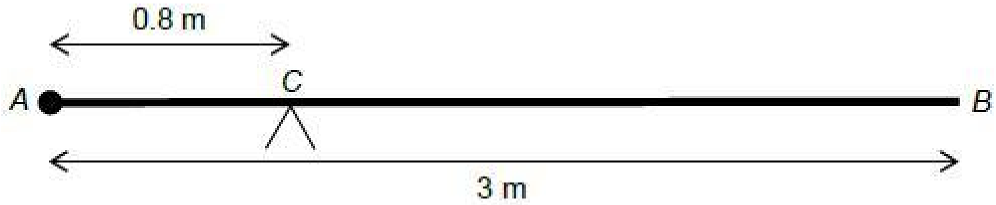
(1)

(Total 9 marks)

Q5.

A uniform rod, *AB*, has length 3 metres and mass 24 kg.

A particle of mass M kg is attached to the rod at *A*.
The rod is balanced in equilibrium on a support at *C*, which is 0.8 metres from *A*.



Find the value of M.

(Total 2 marks)

Q6.

In this question use g = 9.8 m s^{-2}, giving your final answers to an appropriate degree of accuracy.

The diagram shows a box, of mass 8.0 kg, being pulled by a string so that the box moves at a constant speed along a rough horizontal wooden board.

The string is at an angle of 40° to the horizontal.

The tension in the string is 50 newtons.

The coefficient of friction between the box and the board is μ

Model the box as a particle.

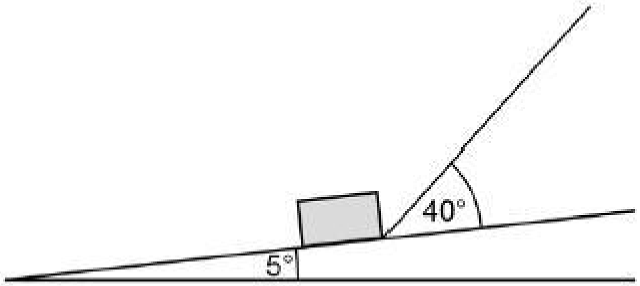
(a) Show that μ = 0.83

(4)

(b) One end of the board is lifted up so that the board is now inclined at an angle of 5° to the horizontal.

The box is pulled up the inclined board.
The string remains at an angle of 40° to the board.

The tension in the string is increased so that the box accelerates up the board at 3 m s^{-2}



(i) Draw a diagram to show the forces acting on the box as it moves.

(1)

(ii) Find the tension in the string as the box accelerates up the slope at 3 m s^{-2}.

(7)

(Total 12 marks)

Q7.

A box, of mass 3 kg, is placed on a rough slope inclined at an angle of 40° to the horizontal. It is released from rest and slides down the slope.

(a) Draw a diagram to show the forces acting on the box.

(1)

(b) Find the magnitude of the normal reaction force acting on the box.

(2)

(c) The coefficient of friction between the box and the slope is 0.2. Find the magnitude of the friction force acting on the box.

(2)

(d) Find the acceleration of the box.

(3)

(e) State an assumption that you have made about the forces acting on the box.

(1)

(Total 9 marks)

Q8.

A tractor, of mass 3500 kg, is used to tow a trailer, of mass 2400 kg, across a horizontal field. The trailer is connected to the tractor by a horizontal tow bar. As they move, a constant resistance force of 800 newtons acts on the trailer and a constant resistance force of R newtons acts on the tractor. A forward driving force of 2500 newtons acts on the tractor. The trailer and tractor accelerate at 0.2 m s^{-2}.

(a) Find R.

(3)

(b) Find the magnitude of the force that the tow bar exerts on the trailer.

(3)

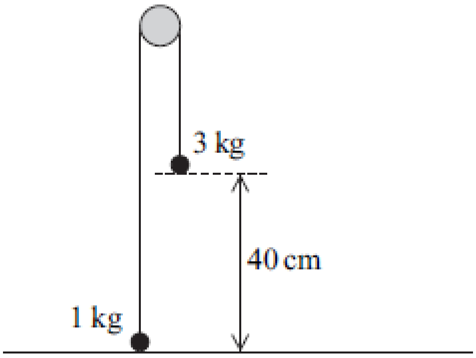
(c) State the magnitude of the force that the tow bar exerts on the tractor.

(1)

(Total 7 marks)

Q9.

Two particles are connected by a light inextensible string that passes over a smooth peg. The particles have masses of 3 kg and 1 kg. The 1 kg particle is pulled down to ground level, where it is 40 cm below the level of the 3 kg particle, as shown in the diagram.



The particles are released from rest with the string vertical above each particle. Assume that no resistance forces act on the particles as they move.

(a) By forming two equations of motion, one for each particle, find the magnitude of the acceleration of the particles after they have been released but before the 3 kg particle hits the ground.

(5)

(b) Find the speed of the 1 kg particle when the 3 kg particle hits the ground.

(2)

(c) After the 3 kg particle has hit the ground, the 1 kg particle continues to move and the string is now slack. Find the maximum height above ground level reached by the 1 kg particle.

(3)

(d) If a constant air resistance force also acts on the particles as they move, explain how this would change your answer for the acceleration in part (a). Give a reason for your answer.

(2)

(Total 12 marks)

Q10.

A block of mass 30 kg is dragged across a rough horizontal surface by a rope that is at an angle of 20° to the horizontal. The coefficient of friction between the block and the surface is 0.4.

(a) The tension in the rope is 150 newtons.

(i) Draw a diagram to show the forces acting on the block as it moves.

(2)

(ii) Show that the magnitude of the normal reaction force on the block is 243 newtons, correct to three significant figures.

(3)

(iii) Find the magnitude of the friction force acting on the block.

(2)

(iv) Find the acceleration of the block.

(4)

(b) When the block is moving, the tension is reduced so that the block moves at a constant speed, with the angle between the rope and the horizontal unchanged. Find the tension in the rope when the block is moving at this constant speed.

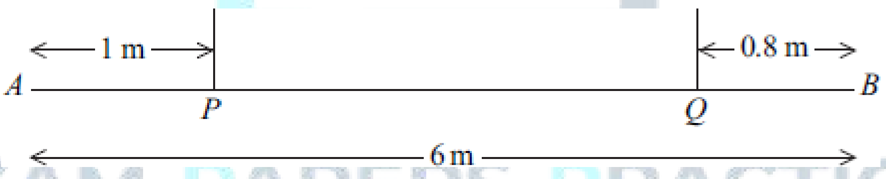
(5)

(c) If the block were made to move at a greater **constant** speed, again with the angle between the rope and the horizontal unchanged, how would the tension in this case compare to the tension found in part (b)?

(1)
(Total 17 marks)

Q11.

A uniform plank AB, of length 6 m, has mass 25 kg. It is supported in equilibrium in a horizontal position by two vertical inextensible ropes. One of the ropes is attached to the plank at the point P and the other rope is attached to the plank at the point Q, where AP = 1 m and QB = 0.8 m, as shown in the diagram.



(a) (i) Find the tension in each rope.

(5)

(ii) State how you have used the fact that the plank is uniform in your solution.

(1)

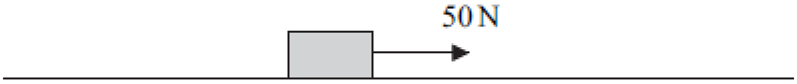
(b) A particle of mass m kg is attached to the plank at point B, and the tension in each rope is now the same.

Find m.

(6)
(Total 12 marks)

Q12.

A block, of mass 4 kg, is made to move in a straight line on a rough horizontal surface by a horizontal force of 50 newtons, as shown in the diagram.



Assume that there is no air resistance acting on the block.

(a) Draw a diagram to show all the forces acting on the block.

(1)

(b) Find the magnitude of the normal reaction force acting on the block.

(1)

(c) The acceleration of the block is 3 m s^{-2}. Find the magnitude of the friction force acting on the block.

(3)

(d) Find the coefficient of friction between the block and the surface.

(2)

(e) Explain how and why your answer to part (d) would change if you assumed that air resistance did act on the block.

(2)

(Total 9 marks)

Q13.

A car, of mass 1200 kg, tows a caravan, of mass 1000 kg, along a straight horizontal road. The caravan is attached to the car by a horizontal towbar. A resistance force of magnitude R newtons acts on the car and a resistance force of magnitude $2R$ newtons acts on the caravan. The car and caravan accelerate at a constant 1.6 m s^{-2} when a driving force of magnitude 4720 newtons acts on the car.

(a) Find R.

(4)

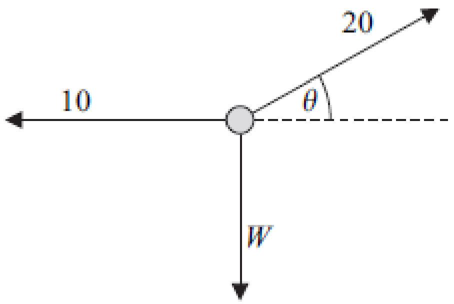
(b) Find the tension in the towbar.

(3)

(Total 7 marks)

Q14.

A particle, of weight W newtons, is held in equilibrium by two forces of magnitudes 10 newtons and 20 newtons. The 10-newton force is horizontal and the 20-newton force acts at an angle θ above the horizontal, as shown in the diagram. All three forces act in the same vertical plane.



(a) Find θ.

(3)

(b) Find W.

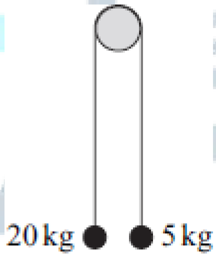
(2)

(c) Calculate the mass of the particle.

(2)
(Total 7 marks)

Q15.

Two particles have masses of 20 kg and 5 kg. They are connected by a light inextensible string that passes over a smooth fixed peg, as shown in the diagram.



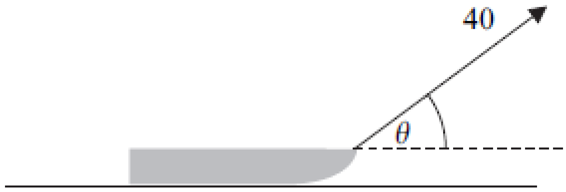
The particles are released from rest.

By forming two equations of motion, find the magnitude of the acceleration of the particles.

(Total 4 marks)

Q16.

A child pulls a sledge, of mass 8 kg, along a rough horizontal surface, using a light rope. The coefficient of friction between the sledge and the surface is 0.3. The tension in the rope is 40 newtons. The rope is kept at an angle of θ to the horizontal, as shown in the diagram.



Model the sledge as a particle.

(a) Draw a diagram to show all the forces acting on the sledge.

(1)

(b) Find the magnitude of the normal reaction force acting on the sledge, in terms of θ.

(3)

(c) The acceleration of the sledge is

$$p + q \cos \theta + r \sin \theta$$

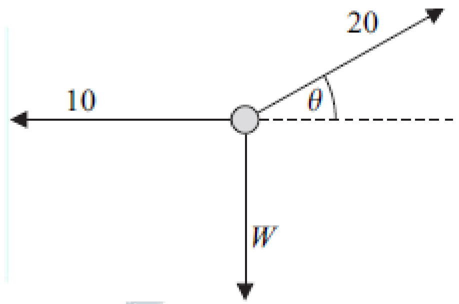
Find p, q and r.

(5)

(Total 9 marks)

Q17.

A particle, of weight W newtons, is held in equilibrium by two forces of magnitudes 10 newtons and 20 newtons. The 10-newton force is horizontal and the 20-newton force acts at an angle θ above the horizontal, as shown in the diagram. All three forces act in the same vertical plane.



(a) Find θ.

(3)

(b) Find W.

(2)

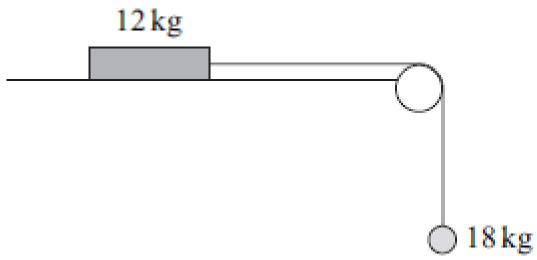
(c) Calculate the mass of the particle.

(2)

(Total 7 marks)

Q18.

A block, of mass 12 kg, lies on a horizontal surface. The block is attached to a particle, of mass 18 kg, by a light inextensible string which passes over a smooth fixed peg. Initially, the block is held at rest so that the string supports the particle, as shown in the diagram.



The block is then released.

(a) Assuming that the surface is smooth, use two equations of motion to find the magnitude of the acceleration of the block and particle.

(4)

(b) In reality, the surface is rough and the acceleration of the block is 3 m s^{-2}.

(i) Find the tension in the string.

(3)

(ii) Calculate the magnitude of the normal reaction force acting on the block.

(1)

(iii) Find the coefficient of friction between the block and the surface.

(5)

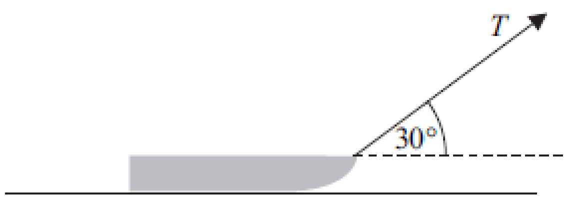
(c) State two modelling assumptions, other than those given, that you have made in answering this question.

(2)

(Total 15 marks)

Q19.

A child pulls a sledge, of mass 8 kg, along a rough horizontal surface, using a light rope. The coefficient of friction between the sledge and the surface is 0.3. The tension in the rope is T newtons. The rope is kept at an angle of 30° to the horizontal, as shown in the diagram.



Model the sledge as a particle.

(a) Draw a diagram to show all the forces acting on the sledge.

(1)

(b) Find the magnitude of the normal reaction force acting on the sledge, in terms of T.

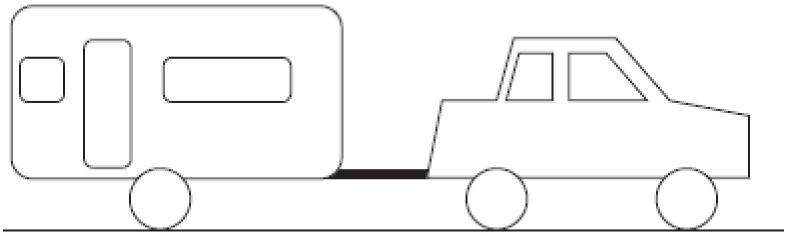
(3)

(c) Given that the sledge accelerates at 0.05 m s^{-2}, find T.

(6)
(Total 10 marks)

Q20.

A car, of mass 1200 kg, tows a caravan, of mass 1000 kg, along a straight horizontal road. The caravan is attached to the car by a horizontal tow bar, as shown in the diagram.



Assume that a constant resistance force of magnitude 200 newtons acts on the car and a constant resistance force of magnitude 300 newtons acts on the caravan.
A constant driving force of magnitude *P* newtons acts on the car in the direction of motion.
The car and caravan accelerate at 0.8 m s^{-2}.

(a) (i) Find *P*.

(3)

(ii) Find the magnitude of the force in the tow bar that connects the car to the caravan.

(3)

(b) (i) Find the time that it takes for the speed of the car and caravan to increase from 7 m s^{-1} to 15 m s^{-1}.

(3)

(ii) Find the distance that they travel in this time.

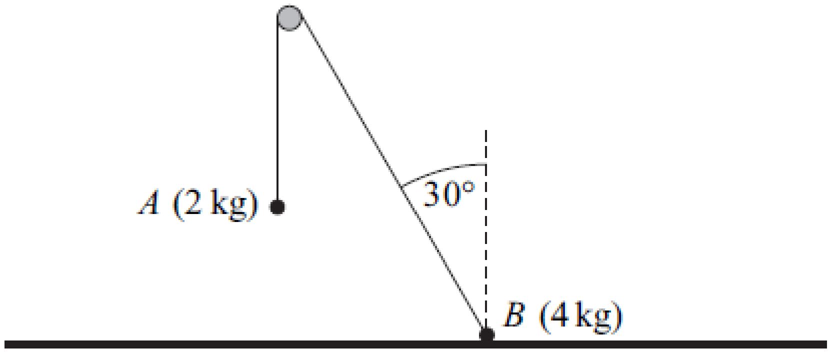
(3)

(c) Explain why the assumption that the resistance forces are constant is unrealistic.

(1)
(Total 13 marks)

Q21.

Two particles, A and B, are connected by a light inextensible string which passes over a smooth peg. Particle A has mass 2 kg and particle B has mass 4 kg. Particle A hangs freely with the string vertical. Particle B is at rest in equilibrium on a rough horizontal surface with the string at an angle of 30° to the vertical. The particles, peg and string are shown in the diagram.



(a) By considering particle A, find the tension in the string.

(2)

(b) Draw a diagram to show the forces acting on particle B.

(2)

(c) Show that the magnitude of the normal reaction force acting on particle B is 22.2 newtons, correct to three significant figures.

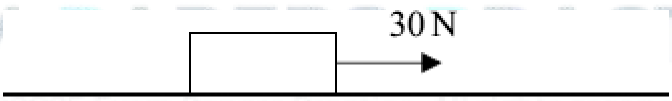
(3)

(d) Find the least possible value of the coefficient of friction between particle B and the surface.

(4)
(Total 11 marks)

Q22.

A wooden block, of mass 4 kg, is placed on a rough horizontal surface. The coefficient of friction between the block and the surface is 0.3 . A horizontal force, of magnitude 30 newtons, acts on the block and causes it to accelerate.



(a) Draw a diagram to show all the forces acting on the block.

(1)

(b) Calculate the magnitude of the normal reaction force acting on the block.

(1)

(c) Find the magnitude of the friction force acting on the block.

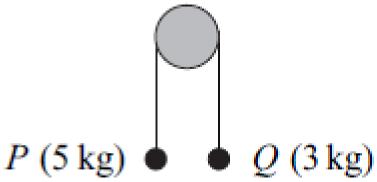
(2)

(d) Find the acceleration of the block.

(3)
(Total 7 marks)

Q23.

Two particles, P and Q, are connected by a string that passes over a fixed smooth peg, as shown in the diagram. The mass of P is 5 kg and the mass of Q is 3 kg.



The particles are released from rest in the position shown.

(a) By forming an equation of motion for each particle, show that the magnitude of the acceleration of each particle is 2.45 m s^{-2}.

(5)

(b) Find the tension in the string.

(2)

(c) State **two** modelling assumptions that you have made about the string.

(2)

(d) Particle P hits the floor when it has moved 0.196 metres and Q has not reached the peg.

(i) Find the time that it takes P to reach the floor.

(3)

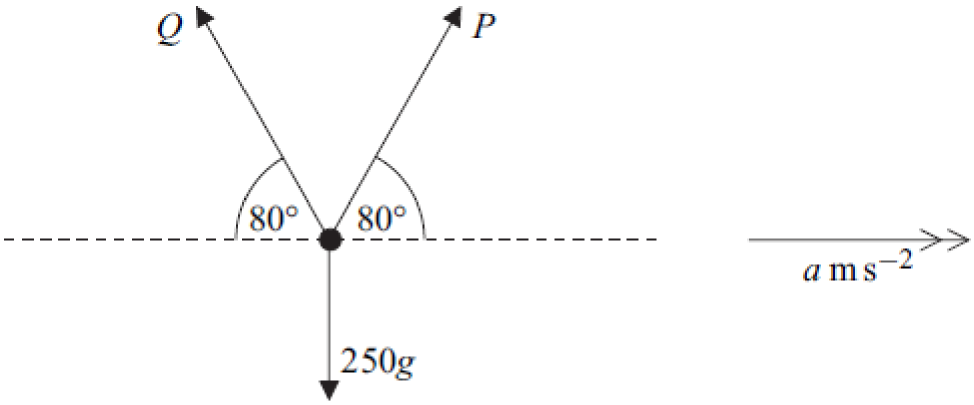
(ii) Find the speed of P when it hits the floor.

(2)

(Total 14 marks)

Q24.

Three forces act in a vertical plane on an object of mass 250 kg, as shown in the diagram.



The two forces P newtons and Q newtons each act at 80° to the horizontal. The object accelerates horizontally at a m s^{-2} under the action of these forces.

(a) Show that

$$P = 125\left(\frac{a}{\cos 80°} + \frac{g}{\sin 80°}\right)$$

(5)

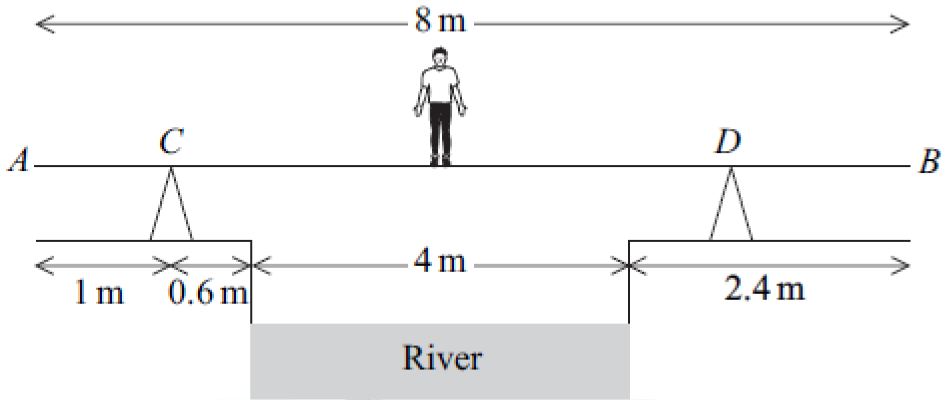
(b) Find the value of a for which Q is zero.

(3)

(Total 8 marks)

Q25.

Ken is trying to cross a river of width 4 m. He has a uniform plank, AB, of length 8 m and mass 17 kg. The ground on both edges of the river bank is horizontal. The plank rests at two points, C and D, on fixed supports which are on opposite sides of the river. The plank is at right angles to both river banks and is horizontal. The distance AC is 1 m, and the point C is at a horizontal distance of 0.6 m from the river bank. Ken, who has mass 65 kg, stands on the plank directly above the middle of the river, as shown in the diagram.



(a) Draw a diagram to show the forces acting on the plank.

(2)

(b) Given that the reaction on the plank at the point D is $44g$ N, find the horizontal distance of the point D from the nearest river bank.

(4)

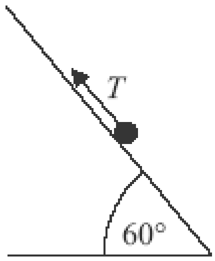
(c) State how you have used the fact that the plank is uniform in your solution.

(1)

(Total 7 marks)

Q26.

A particle of mass 3 kg is on a smooth slope inclined at 60° to the horizontal. The particle is held at rest by a force of T newtons parallel to the slope, as shown in the diagram.



(a) Draw a diagram to show all the forces acting on the particle.

(1)

(b) Show that the magnitude of the normal reaction acting on the particle is 14.7 newtons.

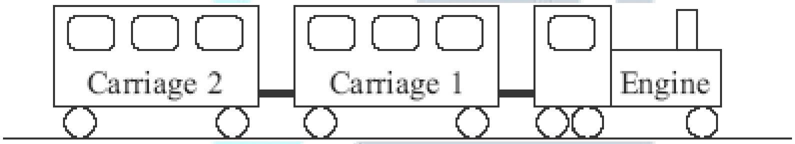
(2)

(c) Find T.

(2)
(Total 5 marks)

Q27.

A small train at an amusement park consists of an engine and two carriages connected to each other by light horizontal rods, as shown in the diagram.



The engine has mass 2000 kg and each carriage has mass 500 kg.

The train moves along a straight horizontal track. A resistance force of magnitude 400 newtons acts on the engine, and resistance forces of magnitude 300 newtons act on each carriage. The train is accelerating at 0.5 m s^{-2}.

(a) Draw a diagram to show the **horizontal** forces acting on Carriage 2.

(1)

(b) Show that the magnitude of the force that the rod exerts on Carriage 2 is 550 newtons.

(2)

(c) Find the magnitude of the force that the rod attached to the engine exerts on Carriage 1.

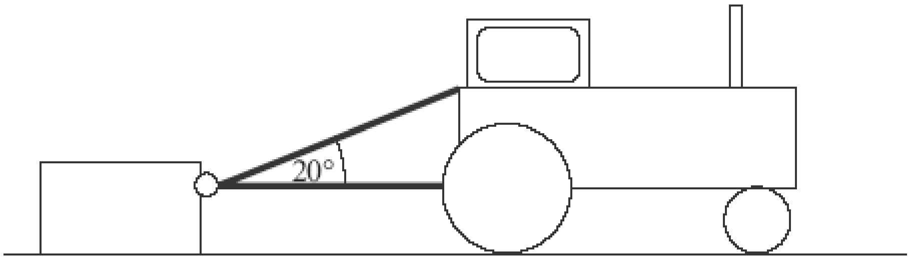
(3)

(d) A forward driving force of magnitude P newtons acts on the engine. Find P.

(3)
(Total 9 marks)

Q28.

A crate, of mass 200 kg, is initially at rest on a rough horizontal surface. A smooth ring is attached to the crate. A light inextensible rope is passed through the ring, and each end of the rope is attached to a tractor. The lower part of the rope is horizontal and the upper part is at an angle of 20° to the horizontal, as shown in the diagram.



When the tractor moves forward, the crate accelerates at 0.3 m s^{-2}. The coefficient of friction between the crate and the surface is 0.4.

Assume that the tension, T newtons, is the same in both parts of the rope.

(a) Draw and label a diagram to show the forces acting on the crate.

(2)

(b) Express the normal reaction between the surface and the crate in terms of T.

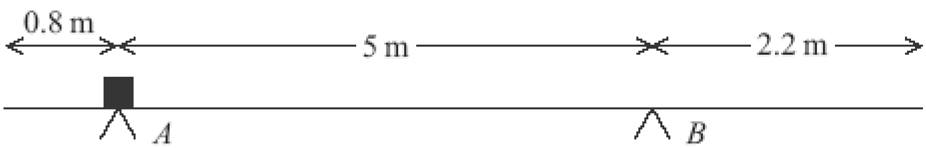
(3)

(c) Find T.

(5)
(Total 10 marks)

Q29.

A uniform plank, of length 8 metres, has mass 30 kg. The plank is supported in equilibrium in a horizontal position by two smooth supports at the points A and B, as shown in the diagram. A block, of mass 20 kg, is placed on the plank at point A.



(a) Draw a diagram to show the forces acting on the plank.

(2)

(b) Show that the magnitude of the force exerted on the plank by the support at B is $19.2g$ newtons.

(3)

(c) Find the magnitude of the force exerted on the plank by the support at A.

(2)

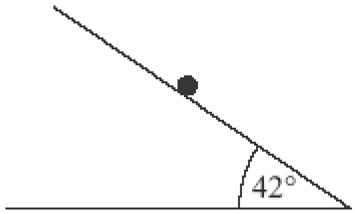
(d) Explain how you have used the fact that the plank is uniform in your solution.

(1)

(Total 8 marks)

Q30.

A particle, of mass m kg, is at rest on a rough plane which is inclined at an angle of 42° to the horizontal, as shown in the diagram.



The friction force acting on the particle has magnitude 30 newtons.

(a) Draw and label a diagram to show all the forces acting on the particle.

(1)

(b) Find m.

(3)

(c) Find the magnitude of the normal reaction force acting on the particle.

(2)

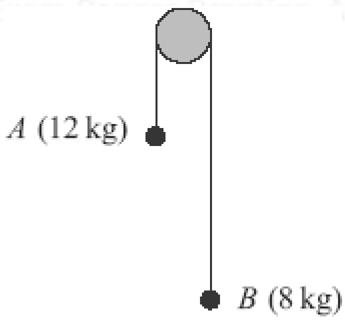
(d) Given that the particle is on the point of sliding down the plane, find the coefficient of friction between the particle and the plane.

(3)

(Total 9 marks)

Q31.

Two particles, A and B, have masses 12 kg and 8 kg respectively. They are connected by a light inextensible string that passes over a smooth fixed peg, as shown in the diagram.



The particles are released from rest and move vertically. Assume that there is no air resistance.

(a) By forming two equations of motion, show that the magnitude of the acceleration of each particle is 1.96 m s^{-2}.

(5)

(b) Find the tension in the string.

(2)

(c) After the particles have been moving for 2 seconds, both particles are at a height of 4 metres above a horizontal surface. When the particles are in this position, the string breaks.

(i) Find the speed of particle A when the string breaks.

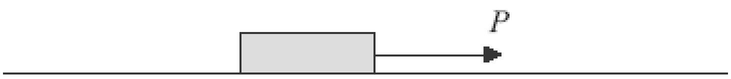
(2)

(ii) Find the speed of particle A when it hits the surface.

(3)
(Total 12 marks)

Q32.

A block, of mass 10 kg, is at rest on a rough horizontal surface, when a horizontal force, of magnitude P newtons, is applied to the block, as shown in the diagram.



The coefficient of friction between the block and the surface is 0.5.

(a) Draw and label a diagram to show all the forces acting on the block.

(1)

(b) (i) Calculate the magnitude of the normal reaction force acting on the block.

(1)

(ii) Find the maximum possible magnitude of the friction force between the block and the surface.

(1)

(iii) Given that P = 30, state the magnitude of the friction force acting on the block.

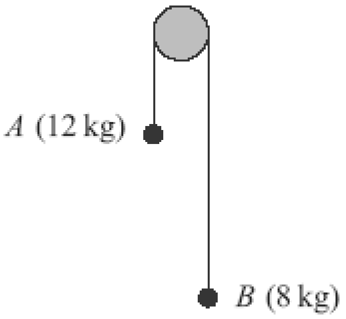
(1)

(c) Given that P = 80, find the acceleration of the block.

(3)
(Total 7 marks)

Q33.

Two particles, A and B, have masses 12 kg and 8 kg respectively. They are connected by a light inextensible string that passes over a smooth fixed peg, as shown in the diagram.



The particles are released from rest and move vertically. Assume that there is no air resistance.

(a) By forming two equations of motion, show that the magnitude of the acceleration of each particle is 1.96 m s^{-2}.

(5)

(b) Find the tension in the string.

(2)

(c) After the particles have been moving for 2 seconds, both particles are at a height of 4 metres above a horizontal surface. When the particles are in this position, the string breaks.

(i) Find the speed of particle A when the string breaks.

(2)

(ii) Find the speed of particle A when it hits the surface.

(3)

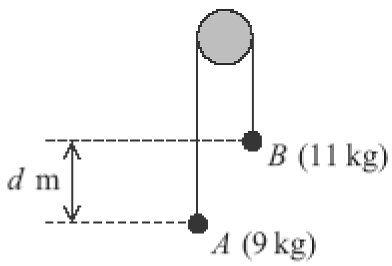
(iii) Find the time that it takes for particle B to reach the surface after the string breaks.
Assume that particle B does not hit the peg.

(5)

(Total 17 marks)

Q34.

Two particles, A and B, are connected by a light inextensible string that passes over a smooth fixed peg, as shown in the diagram. The mass of A is 9 kg and the mass of B is 11 kg.



The particles are released from rest in the position shown, where B is d metres higher

than A.
Assume that no resistance forces act on the particles.

(a) By forming an equation of motion for each of the particles A and B, show that the acceleration of each particle has magnitude 0.98 m s^{-2}.

(5)

(b) When the particles have been moving for 0.5 seconds, they are at the same level.

(i) Find the speed of the particles at this time.

(2)

(ii) Find d.

(4)

(Total 11 marks)

Q35.

A box of mass 4 kg is held at rest on a plane inclined at an angle of 40° to the horizontal. The box is then released and slides down the plane.

(a) A simple model assumes that the only forces acting on the box are its weight and the normal reaction from the plane. Show that, according to this simple model, the acceleration of the box would be 6.30 m s^{-2}, correct to three significant figures.

(3)

(b) In fact, the box moves down the plane with constant acceleration and travels 0.9 metres in 0.6 seconds. By using this information, find the acceleration of the box.

(3)

(c) Explain why the answer to part (b) is less than the answer to part (a).

(1)

(Total 7 marks)